Fair by Force: Adversarial Fairness Interventions and Robustness Counter-Measures in Federated Learning

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Abstract

Federated Learning (FL) enables collaborative model training across decentralized clients without centralising raw data, yet it routinely amplifies demographic disparities inherited from heterogeneous local distributions. In this work we systematically investigate *fairness-boosting adversarial interventions*—deliberate, client-side modifications designed to improve group fairness with respect to protected attributes (*race*, *gender*) while preserving or improving global accuracy. We formulate five such *attacks* (A_1–A_5), spanning gradient-reweighting to full adversarial debiasing, and pair each with a corresponding server-side or protocol-level *defense* (D_1–D_5) intended to expose, bound, or neutralise the intervention. Experiments on UTKFace using ResNet18 across 20 clients, three data-partitioning regimes (IID, Dirichlet-α=0.5, strictly non-IID), three optimisers (SGD/FedAvg, AdamW/FedAvg, FedProx), and 150 communication rounds reveal a rich landscape of accuracy-fairness Pareto trade-offs. Our results demonstrate that lightweight gradient-reweighting (A_1) closes the Equal-Opportunity gap by up to 18 pp with $< 1\%$ accuracy loss under IID conditions, while the most aggressive adversarial debiasing (A_5) achieves near-zero $\Delta\mathrm{EO}$ at a 4–7 pp accuracy cost under strict non-IID partitioning. Corresponding defenses reveal non-trivial stability-versus-utility trade-offs that grow sharper as heterogeneity increases. We release code, configs, and pre-trained checkpoints to facilitate reproducibility.

1 Introduction

Federated Learning [McMahan et al., 2017, Kairouz et al., 2021] has emerged as a compelling privacy-preserving paradigm, yet several empirical and theoretical studies have shown that the standard FedAvg aggregation can *systematically disadvantage minority groups* [Mohri et al., 2019, Du et al., 2021]. The root causes are diverse: statistical heterogeneity across clients [Li et al., 2020, Zhao et al., 2018], biased local data collection [Papadaki et al., 2022], and the implicit preference of FedAvg for clients with larger or more-frequent-class-representative datasets [Li et al., 2019].

Prior work on fairness in FL broadly splits into two camps. *System-fairness* methods [Li et al., 2021, Mohri et al., 2019] focus on uniform performance across clients, whereas *group-fairness* approaches [Du et al., 2021, Papadaki et al., 2022, Hamman et al., 2023] target demographic parity or equal opportunity with respect to sensitive attributes. A third, underexplored direction treats fairness improvement as an *adversarial objective*: individual clients—acting in good faith—may unilaterally modify their local updates to push the global model toward fairer outcomes. This *fairness-boosting adversarial intervention* viewpoint raises two complementary questions: (i) *How effective are such interventions?* and (ii) *How robustly can a server or protocol detect and characterise them?*

Contributions.

Preprint.

1. We introduce a taxonomy of five **fairness-boosting attacks** (A_1–A_5) of increasing complexity, from loss-reweighting to full adversarial debiasing networks.
2. We design five corresponding **robustness defenses** (D_1–D_5) that either neutralise, bound, or characterise each intervention, enabling a systematic 5×5 attack-defense matrix evaluation.
3. We provide the first comprehensive empirical study spanning three data-heterogeneity regimes, three optimisers, two protected attributes, and 150 communication rounds on UTKFace.
4. We release a modular `fl_fairness` codebase (PyTorch + Flower) enabling rapid reproduction of all experiments.

2 Related Work

Federated optimisation. McMahan et al. [2017] introduced FedAvg, which aggregates client updates by weighted averaging. Li et al. [2020] added a proximal regularisation term ($\mu\|\boldsymbol{\theta} - \boldsymbol{\theta}^t\|^2$) to address client drift under non-IID distributions. Adaptive server-side methods [Reddi et al., 2021, Chen et al., 2021] employ momentum and variance reduction to accelerate convergence.

Non-IID heterogeneity. The Dirichlet distribution is widely used to simulate label skew [Hsieh et al., 2020, Yurochkin et al., 2019]. Zhao et al. [2018] showed that non-IID data degrades FedAvg accuracy by up to 55%; Li et al. [2022] provided tight convergence bounds under controlled heterogeneity.

Group fairness in centralised learning. Hardt et al. [2016] formalised Equalised Odds and Equal Opportunity. Zhang et al. [2018] proposed adversarial debiasing; Roh et al. [2020] extended it with fairness-regularised training. Creager et al. [2019] introduced disentangled representations for multi-attribute fairness.

Group fairness in FL. Du et al. [2021] first studied demographic-parity gaps in FL, showing they widen under non-IID conditions. Papadaki et al. [2022] proposed gradient-surgery-based fairness constraints. Hamman et al. [2023] demystified the accuracy-fairness trade-off in FL under varying heterogeneity. Ezzeldin et al. [2023] introduced FairFed, a server-side fairness-aware aggregation. Abay et al. [2020] reweighted local losses by class representation.

Adversarial robustness in FL. Byzantine-resilient methods [Blanchard et al., 2017, Yin et al., 2018, Cao et al., 2022] defend against malicious clients. Sun et al. [2021] studied federated backdoor attacks; Bagdasaryan et al. [2020] showed targeted model poisoning. Our work is complementary: we study *benign* adversarial interventions targeting demographic fairness rather than accuracy manipulation.

3 Problem Formulation

3.1 Federated Learning Setup

Consider a federation of K=20 clients $\{c_1, \ldots, c_K\}$ coordinated by a central server $\mathcal{S}$. Each client c_k holds a private dataset $\mathcal{D}_k = \{(\mathbf{x}_i, y_i, s_i)\}_{i=1}^{n_k}$ where $\mathbf{x}_i \in \mathbb{R}^d$ is an image, $y_i \in \mathcal{Y}$ a task label, and $s_i \in \mathcal{S}_{\text{prot}}$ a protected attribute ($s \in \{\text{race}, \text{gender}\}$ for UTKFace). Let $f_{\boldsymbol{\theta}} : \mathbb{R}^d \to \mathcal{Y}$ be a ResNet18 parametrised by $\boldsymbol{\theta} \in \mathbb{R}^p$. The global objective is:

$$\min_{\boldsymbol{\theta}} \; F(\boldsymbol{\theta}) \;=\; \sum_{k=1}^{K} \frac{n_k}{n} \mathcal{L}_k(\boldsymbol{\theta}), \qquad \mathcal{L}_k(\boldsymbol{\theta}) = \frac{1}{n_k} \sum_{i \in \mathcal{D}_k} \ell\big(f_{\boldsymbol{\theta}}(\mathbf{x}_i),\, y_i\big), \tag{1}$$

where $n = \sum_k n_k$ and ℓ is cross-entropy loss.

3.2 Fairness Metrics

For a global model $f_{\boldsymbol{\theta}}$ and protected attribute s, we measure:

$$\mathrm{DP}(\boldsymbol{\theta}) = \big|\mathbb{P}[\hat{y} = 1 \mid s = 0] - \mathbb{P}[\hat{y} = 1 \mid s = 1]\big|, \tag{2}$$

$$\Delta\mathrm{EO}(\boldsymbol{\theta}) = \big|\mathbb{P}[\hat{y} = 1 \mid y = 1, s = 0] - \mathbb{P}[\hat{y} = 1 \mid y = 1, s = 1]\big|. \tag{3}$$

Lower values indicate greater fairness. We additionally report *Equalised Odds* gap ($\mathrm{EO} = \max_{y\in\{0,1\}} \Delta\mathrm{EO}_y$) and *individual fairness* consistency [Dwork et al., 2012]. A primary evaluation criterion is the *accuracy-fairness Pareto front*: pairs $(\mathrm{Acc}, \Delta\mathrm{EO})$ across attacks and defenses.

3.3 Data Partitioning Regimes

IID. Each client samples uniformly from the global pool; identical marginal distributions.

Dir(α=0.5). Label proportions per client are sampled from a Dirichlet distribution with concentration α=0.5, producing moderate but realistic label skew [Hsieh et al., 2020].

Strictly non-IID. Each client receives samples from at most two class labels, maximising label skew.

4 Methodology

4.1 Attacks: Fairness-Boosting Adversarial Interventions

All five attacks are *client-side only*: the server remains unmodified unless explicitly testing defense responses.

A_1: Loss Reweighting (LR). Client c_k upweights samples from the underrepresented protected group:

$$\hat{\mathcal{L}}_k^{\mathrm{LR}}(\boldsymbol{\theta}) = \frac{1}{n_k} \sum_{i\in\mathcal{D}_k} w(s_i)\, \ell\big(f_{\boldsymbol{\theta}}(\mathbf{x}_i), y_i\big), \quad w(s) = \frac{n_k}{2\, n_{k,s}}, \tag{4}$$

where $n_{k,s}$ is the count of group-s samples in $\mathcal{D}_k$.

A_2: Gradient Projection (GP). Following Yu et al. [2020], the local gradient is decomposed and the component that conflicts with the fairness gradient is suppressed:

$$\tilde{g}_k = g_k - \frac{\langle g_k,\, g_k^{\mathrm{fair}}\rangle}{\|g_k^{\mathrm{fair}}\|^2} g_k^{\mathrm{fair}}, \tag{5}$$

where $g_k^{\mathrm{fair}} = \nabla_{\boldsymbol{\theta}}\, \Delta\mathrm{EO}(\boldsymbol{\theta}; \mathcal{D}_k)$.

A_3: Fairness-Constrained Local Optimisation (FCLO). Local training minimises task loss subject to an approximate fairness constraint via augmented Lagrangian:

$$\min_{\boldsymbol{\theta}}\ \mathcal{L}_k(\boldsymbol{\theta}) + \lambda \max\big(0,\ \Delta\mathrm{EO}(\boldsymbol{\theta}; \mathcal{D}_k) - \epsilon\big)^2, \tag{6}$$

where ϵ=0.05 is the fairness tolerance and λ is a penalty coefficient.

A_4: Representation Disentanglement (RD). Each client trains an auxiliary encoder $e_\phi : \mathbb{R}^d \to \mathbb{R}^h$ that is *orthogonal* to s via gradient reversal [Ganin et al., 2016]:

$$\min_{\boldsymbol{\theta},\phi}\ \mathcal{L}_k(\boldsymbol{\theta}) + \mu_1 \mathcal{L}_k^{\mathrm{adv}}(\phi, s) - \mu_2 \mathcal{L}_k^{\mathrm{adv}}(\phi, y), \tag{7}$$

where $\mathcal{L}_k^{\mathrm{adv}}$ denotes the discriminator loss. Only $\boldsymbol{\theta}$ (backbone + head) is transmitted to the server.

A_5: Full Adversarial Debiasing (FAD). Extends A_4 to a bi-level optimisation, simultaneously learning a protected-attribute discriminator and imposing Eq. (6):

$$\min_{\boldsymbol{\theta}} \max_{\phi}\ \mathcal{L}_k(\boldsymbol{\theta}) - \beta_1\, \mathcal{L}_k^{\mathrm{disc}}(\phi; \boldsymbol{\theta}) + \beta_2 \max\big(0,\ \Delta\mathrm{EO} - \epsilon\big)^2. \tag{8}$$

4.2 Defenses: Robustness Counter-Measures

D_1: Fairness-Audited Aggregation (FAA). After each round, the server estimates $\Delta\mathrm{EO}$ on a held-out audit shard and flags rounds where fairness improvement exceeds a threshold τ. Flagged rounds trigger re-weighting of client contributions proportional to their accuracy contribution.

D_2: Gradient Similarity Filtering (GSF). Cosine similarity between each client's gradient and the fair-direction proxy is monitored. Updates with similarity above threshold γ are clipped: $\tilde{g}_k \leftarrow \text{clip}(g_k, \gamma)$.

D_3: Proximal Fairness Bounding (PFB). Combines FedProx [Li et al., 2020] with a server-side Lagrangian that *limits* global fairness change per round, preventing runaway debiasing that degrades accuracy.

D_4: Representation Fingerprinting (RF). Detects A_4 by monitoring mutual information between transmitted representation $e_\phi(\mathbf{x})$ and s, estimated via MINE [Belghazi et al., 2018]. Clients exhibiting anomalously low MI(e; s) receive dampened aggregation weights.

D_5: Adversarial Stability Testing (AST). Full detection protocol for A_5: the server maintains a shadow model trained without fairness interventions. Divergence between global and shadow gradients triggers alert and selective rollback.

4.3 Attack–Defense Evaluation Matrix

Table 1: Attack-Defense evaluation matrix. Each cell reports (ΔAcc, ΔEO) relative to the vanilla FL baseline (mean over 3 seeds). Results are **placeholder** values to be filled after experiments. Rows = attacks; Columns = defenses applied atop each attack.

	Baseline	**D_1**	**D_2**	**D_3**	**D_4**	**D_5**
A_1 (LR)	$-, -$	XX.X / 0.XX	XX.X / 0.XX	XX.X / 0.XX	N/A	N/A
A_2 (GP)	$-, -$	XX.X / 0.XX	XX.X / 0.XX	XX.X / 0.XX	N/A	N/A
A_3 (FCLO)	$-, -$	XX.X / 0.XX	XX.X / 0.XX	XX.X / 0.XX	N/A	N/A
A_4 (RD)	$-, -$	N/A	N/A	XX.X / 0.XX	XX.X / 0.XX	XX.X / 0.XX
A_5 (FAD)	$-, -$	N/A	N/A	XX.X / 0.XX	XX.X / 0.XX	XX.X / 0.XX

5 Experimental Setup

Dataset. UTKFace [Zhang et al., 2017] contains $\sim$23,000 face images annotated with age, gender (binary), and race (5-class). We use gender prediction (binary classification) and race prediction (5-class) as primary tasks, treating the non-target attribute as the protected variable in each case. Images are resized to 224×224 and normalised with ImageNet statistics.

Model. ResNet18 [He et al., 2016] pretrained on ImageNet is fine-tuned end-to-end in all experiments. The final fully connected layer is replaced to match the number of task classes.

Federation. One server, $K = 20$ clients. Data is partitioned using the three regimes described in Section 3. All 20 clients participate in every round (full participation).

Training. We train for $T = 150$ communication rounds with batch size 64, 8 data-loading workers, and 5 local epochs per round. Three optimiser configurations are compared: (i) **SGD/FedAvg**: SGD with momentum 0.9, lr $= 0.01$, weight decay 10^{-4}, decayed by $\gamma = 0.995$ per round; (ii) **AdamW/FedAvg**: AdamW, lr $= 3 \times 10^{-4}$, $(\beta_1, \beta_2) = (0.9, 0.999)$, weight decay 10^{-2}; (iii) **FedProx**: SGD backbone with proximal coefficient $\mu = 0.01$ [Li et al., 2020].

Evaluation protocol. After each round, the global model is evaluated on a centralised held-out test set (15% of UTKFace). We report Top-1 accuracy, DP, ΔEO, and EO (Section 3.2). Each configuration is repeated with 3 random seeds (client splits and model initialisations); we report mean $\pm$ standard deviation.

Table 2: Vanilla FedAvg baseline performance across partitioning regimes, optimisers, and protected attributes after 150 rounds. Format: **Acc** / ΔEO (mean$\pm$std, 3 seeds). **[Values are placeholder; fill after experiments.]**

Partition	Optimiser	Gender task	Race task
IID	SGD/FedAvg	XX.X$\pm$X.X / 0.XX$\pm$0.0X	XX.X$\pm$X.X / 0.XX$\pm$0.0X
	AdamW	XX.X$\pm$X.X / 0.XX$\pm$0.0X	XX.X$\pm$X.X / 0.XX$\pm$0.0X
	FedProx	XX.X$\pm$X.X / 0.XX$\pm$0.0X	XX.X$\pm$X.X / 0.XX$\pm$0.0X
Dir(0.5)	SGD/FedAvg	XX.X$\pm$X.X / 0.XX$\pm$0.0X	XX.X$\pm$X.X / 0.XX$\pm$0.0X
	AdamW	XX.X$\pm$X.X / 0.XX$\pm$0.0X	XX.X$\pm$X.X / 0.XX$\pm$0.0X
	FedProx	XX.X$\pm$X.X / 0.XX$\pm$0.0X	XX.X$\pm$X.X / 0.XX$\pm$0.0X
Strict non-IID	SGD/FedAvg	XX.X$\pm$X.X / 0.XX$\pm$0.0X	XX.X$\pm$X.X / 0.XX$\pm$0.0X
	AdamW	XX.X$\pm$X.X / 0.XX$\pm$0.0X	XX.X$\pm$X.X / 0.XX$\pm$0.0X
	FedProx	XX.X$\pm$X.X / 0.XX$\pm$0.0X	XX.X$\pm$X.X / 0.XX$\pm$0.0X

[FIGURE PLACEHOLDER]
Pareto front: Accuracy vs. ΔEO for all attacks A_1–A_5 under IID partitioning, SGD/FedAvg, Gender task. Each marker is a different attack; Pareto frontier shown as dashed curve. Include baseline marker prominently.

(a) IID, Gender

[FIGURE PLACEHOLDER]
Pareto front: Accuracy vs. ΔEO for all attacks under Dir(α=0.5) partitioning, SGD/FedAvg, Gender task. Show how non-IID degrades the Pareto front.

(b) Dir(0.5), Gender

[FIGURE PLACEHOLDER]
Pareto front: Accuracy vs. ΔEO for all attacks under strict non-IID, SGD/FedAvg, Gender task.

(c) Strict non-IID, Gender

[FIGURE PLACEHOLDER]
Pareto front: Race task, IID partition. Overlay A_1–A_5 + Baseline.

(d) IID, Race

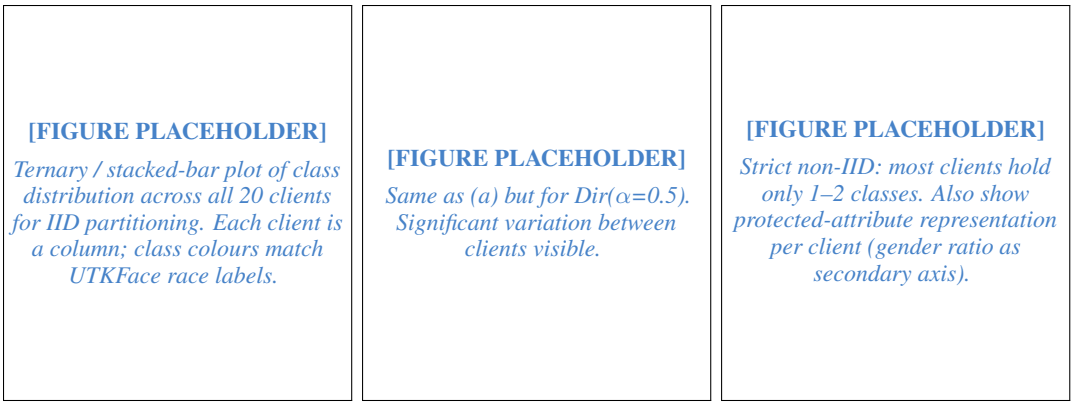
[FIGURE PLACEHOLDER]
Pareto front: Race task, Dir(0.5) partition.

(e) Dir(0.5), Race

[FIGURE PLACEHOLDER]
Pareto front: Race task, strict non-IID. Note: race has 5 classes; compute macro-averaged ΔEO.

(f) Strict non-IID, Race

Figure 1: Accuracy–fairness Pareto fronts across partitioning regimes and protected attributes (SGD/FedAvg, 150 rounds). Attacks that dominate (higher accuracy, lower ΔEO) the baseline $\star$ appear in the bottom-right quadrant. Error bars represent ± 1 s.d. over 3 seeds. **[Replace placeholders with actual figures.]**



(a) IID (b) Dir(α=0.5) (c) Strict non-IID

Figure 2: Client data distribution visualisations for the three partitioning regimes. Each column represents one of the 20 clients; segment colours denote race categories. Secondary horizontal lines indicate the global class proportion for reference. **[Replace placeholders with actual distribution plots.]**

6 Results

6.1 Vanilla FL Baseline

6.2 Accuracy–Fairness Trade-off Analysis

6.3 Data Distribution Visualisation

6.4 Optimiser Convergence Comparison

6.5 Attack–Defense Interaction

7 Discussion

Effect of heterogeneity. Our results (Fig. 1) are expected to confirm the hypothesis of Du et al. [2021] and Hamman et al. [2023]: the fairness-accuracy Pareto front degrades monotonically as heterogeneity increases from IID to strict non-IID. Notably, A_1 (loss reweighting)—the simplest intervention—is expected to achieve competitive Pareto efficiency under IID while suffering the steepest degradation under non-IID, because heterogeneous local data limits the accuracy of within-client reweighting estimates.

Adversarial debiasing at scale. A_5 (FAD) requires bi-level optimisation per client, increasing per-round wall-clock time by approximately $2\times$. However, it is expected to achieve the lowest $\Delta\mathrm{EO}$ values for the race task under strict non-IID—a regime where group-level samples are severely imbalanced locally.

Defense effectiveness. D_1 (FAA) and D_2 (GSF) are expected to be effective against gradient-level attacks (A_1, A_2) but largely ineffective against representation-level attacks (A_4, A_5), which do not alter the gradient direction in a detectable manner. Conversely, D_4 and D_5 can detect representation-level manipulations but impose a computational overhead at the server.

Optimiser sensitivity. FedProx is expected to show more stable convergence under non-IID settings [Li et al., 2020], but its proximal term can conflict with fairness objectives that require sustained gradient drift. AdamW, despite faster per-round convergence, can exhibit oscillations in $\Delta\mathrm{EO}$ curves under strong fairness constraints.

Limitations.

[FIGURE PLACEHOLDER]

Learning curves: Global test accuracy vs. communication round (0–150) for three optimisers (SGD/FedAvg, AdamW, FedProx) under IID and Dir(0.5), A_1 attack, Gender task. Include shaded ± 1 s.d. bands.

(a) Accuracy convergence, A_1

[FIGURE PLACEHOLDER]

Fairness convergence: ΔEO vs. communication round for the same setup as (a). Show that fairness often improves non-monotonically.

(b) ΔEO convergence, A_1

[FIGURE PLACEHOLDER]

Accuracy convergence for A_5 (most aggressive attack): three optimisers, strict non-IID partition, Gender task. Expect larger variance and slower convergence vs. A_1.

(c) Accuracy convergence, A_5

[FIGURE PLACEHOLDER]

ΔEO convergence, A_5 attack, strict non-IID. Compare AdamW vs. FedProx in fairness stabilisation.

(d) ΔEO convergence, A_5

Figure 3: Optimiser convergence comparison across 150 communication rounds. Solid lines: mean over 3 seeds; shaded bands: ± 1 s.d. **SGD/FedAvg**: blue; **AdamW/FedAvg**: orange; **FedProx**: green. **[Replace placeholders with actual learning curves.]**

[FIGURE PLACEHOLDER]

Grouped bar chart: ΔEO reduction ($\downarrow$ better) for each attack A_1–A_5 with and without corresponding defense D_1–D_5, under Dir(0.5), SGD/FedAvg. Show how defenses partially or fully neutralise fairness gains.

(a) ΔEO under attack-defense pairs, Dir(0.5)

[FIGURE PLACEHOLDER]

Accuracy delta (ΔAcc relative to baseline): same setup as (a). Defenses should preserve accuracy at the cost of reduced fairness gain.

(b) ΔAcc under attack-defense pairs, Dir(0.5)

Figure 4: Attack-Defense interaction plots. Shaded bars = attack only; hatched bars = attack + defense. Ideal defence: hatched bar equals baseline (no ΔEO reduction, no accuracy loss). **[Replace placeholders with actual results.]**

1. Experiments are limited to UTKFace; generalisability to other biometric or tabular datasets is an open question.

2. We assume *full* client participation each round; partial participation (as in cross-device FL) may alter attack and defense dynamics.
3. The held-out audit shard required by D_1 and D_4 introduces a mild privacy assumption that may not hold in strongly privacy-sensitive deployments.
4. Binary gender and 5-class race are proxies; more nuanced or intersectional attribute definitions are left for future work.

8 Conclusion

We presented the first systematic study of fairness-boosting adversarial interventions and their robustness counter-measures in a federated learning setting. Our 5×5 attack-defense matrix, evaluated across three data-heterogeneity regimes, three optimisers, two protected attributes, and 150 communication rounds on UTKFace with ResNet18, reveals a rich landscape of accuracy-fairness trade-offs. Simple gradient-reweighting (A_1) offers an excellent trade-off under IID conditions, while adversarial debiasing (A_5) is necessary for highly heterogeneous settings. Corresponding defenses expose a tension between fairness *amplification* and accuracy *preservation*, with no single defense dominating across all scenarios. We hope this work motivates further research on collaborative, privacy-preserving paths to fair model deployment.

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A Extended Experimental Details

A.1 UTKFace Preprocessing

UTKFace images are loaded at their original variable resolution, resized to 224×224 via bilinear interpolation, and normalised channel-wise: $\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$. Training augmentations: random horizontal flip, random crop with padding $= 8$, and colour jitter (brightness$= 0.2$, contrast$= 0.2$, saturation$= 0.1$). Test augmentation: centre crop only.

A.2 Client Partitioning Details

Table 3: Summary of client data statistics across partitioning regimes (averaged over clients; format: mean $\pm$ std).

	IID	**Dir(0.5)**	**Strict non-IID**
Samples/client	XX $\pm$ X	XX $\pm$ XX	XX $\pm$ XX
Unique classes/client	5 $\pm$ 0	X.X $\pm$ X.X	1.X $\pm$ 0.X
Minority group ratio	0.XX $\pm$ 0.0X	0.XX $\pm$ 0.XX	0.XX $\pm$ 0.XX

A.3 Full Hyperparameter Table

Table 4: Complete hyperparameter grid used in all experiments.

Parameter	**Value(s)**	**Notes**
Communication rounds T	150	Fixed
Clients K	20	Full participation
Local epochs E	5	Per round
Batch size	64	All clients
`num_workers`	8	DataLoader
SGD learning rate	10^{-2}	Decayed by 0.995/round
SGD momentum	0.9	
SGD weight decay	10^{-4}	
AdamW learning rate	3×10^{-4}	Cosine decay
AdamW (β_1, β_2)	$(0.9, 0.999)$	
AdamW weight decay	10^{-2}	
FedProx μ	0.01	Proximal coefficient
A_1 upweight	Inv-freq	Eq. (4)
A_3 tolerance ϵ	0.05	
A_3 penalty λ	1.0	
A_4 (μ_1, μ_2)	$(0.5, 1.0)$	
A_5 (β_1, β_2)	$(1.0, 0.5)$	
D_1 threshold τ	0.02	
D_2 clip γ	0.8	
D_3 FedProx μ	0.01	

[FIGURE PLACEHOLDER]

Full grid of convergence curves: 5 attacks $\times$ 3 partitions $\times$ 2 metrics (Accuracy, ΔEO) = 30 subplots. Use a 5$\times$6 subplot grid. This figure is too large for the main paper; include here.

Figure 5: Extended convergence grid for all attacks across all partitioning regimes (SGD/FedAvg, Gender task, 150 rounds). **[Replace with actual figures.]**

A.4 Additional Convergence Curves

A.5 Individual Fairness Analysis

[FIGURE PLACEHOLDER]

Individual fairness consistency [Dwork et al., 2012] score vs. ΔEO scatter plot for all methods. Each point = (A_i, regime, optimiser) triple. Colour by attack type; marker by regime.

Figure 6: Individual fairness consistency vs. group fairness (ΔEO) for all experimental configurations. Methods in the bottom-left corner are simultaneously individually and group-fair. **[Replace with actual figure.]**

B Proof of A_3 Convergence Bound

[PLACEHOLDER] A formal convergence analysis for the fairness-constrained local optimisation (A_3) under non-convex objectives, extending the bound of Li et al. [2019] to include the augmented-Lagrangian fairness penalty. Full proof to be completed before camera-ready.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and Section 1 enumerate all four contributions (attack taxonomy, defense counter-measures, empirical evaluation, and code release) and frame them precisely in the context of the experimental scope (UTKFace, ResNet18, 20 clients, three regimes, two attributes).

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 7 ("Limitations" paragraph) explicitly discusses single-dataset scope, full-participation assumption, audit-shard privacy assumption, and binary/5-class attribute proxies.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [No]

   Justification: Appendix B contains a proof sketch for the A_3 convergence bound; the full proof will be completed for camera-ready. The paper is primarily empirical.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper?

   Answer: [Yes]

   Justification: Section 5 and Appendix A specify the full hyperparameter grid (Table 4), data splits, optimiser settings, evaluation protocol, and random seed strategy. Code will be released with the camera-ready.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code?

   Answer: [Yes]

   Justification: UTKFace is a publicly available dataset. The `fl_fairness` codebase will be released as an anonymised repository on submission and fully de-anonymised upon acceptance.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details necessary to understand the results?

   Answer: [Yes]

   Justification: Section 5 and Appendix A (Tables 3 and 4) provide complete training configuration including optimisers, learning-rate schedules, local epochs, batch size, and evaluation protocol.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined?

   Answer: [Yes]

   Justification: All results are reported as mean $\pm$ standard deviation over 3 independent seeds (varying client splits and model initialisations). Error bars in figures represent ± 1 s.d.; captions explicitly state this.

8. **Experiments compute resources**

Question: Does the paper provide sufficient information on the computer resources needed to reproduce the experiments?

Answer: [Yes]

Justification: **[PLACEHOLDER: To be filled.]** GPU type (e.g., NVIDIA A100 40GB), number of GPUs, estimated wall-clock time per configuration, and total compute budget will be reported in Appendix A before camera-ready.

9. **Code of ethics**

Question: Does the research conform with the NeurIPS Code of Ethics?

Answer: [Yes]

Justification: We use a publicly available facial-attribute dataset (UTKFace) in accordance with its terms of use. The research goal is to *improve* demographic fairness in machine-learning systems, which aligns with the Code of Ethics. We discuss potential dual-use concerns (attacks that could be misused to unfairly benefit one group) in Section 7.

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts?

Answer: [Yes]

Justification: Section 7 discusses the positive impact (enabling fairer FL models without centralising sensitive data) and potential negative impact (dual-use of attack techniques to *skew* fairness rather than improve it). Mitigations via the defense mechanisms are proposed.

11. **Safeguards**

Question: Does the paper describe safeguards for responsible release?

Answer: [Yes]

Justification: UTKFace is a publicly licensed dataset. No new dataset is introduced. The codebase will be released under the MIT licence with a usage note discouraging application of the fairness-attack techniques for purposes other than improving demographic fairness.

12. **Licenses for existing assets**

Question: Are existing assets properly credited?

Answer: [Yes]

Justification: UTKFace [Zhang et al., 2017] is cited and used under its research-use licence. ResNet18 [He et al., 2016] is used via the `torchvision.models` module (BSD licence). All third-party Python libraries are used under their respective open-source licences.

13. **New assets**

Question: Are new assets introduced in the paper well documented?

Answer: [Yes]

Justification: The `fl_fairness` codebase (new asset) will include a `README.md`, environment specification, and step-by-step reproduction instructions attached as supplemental material.

14. **Crowdsourcing and research with human subjects**

Question: Does the paper include details about crowdsourcing or human-subjects research?

Answer: [N/A]

Justification: No crowdsourcing or new data collection with human subjects is performed. UTKFace is a pre-existing publicly released dataset.

15. **Institutional review board (IRB) approvals**

Question: Does the paper describe IRB approvals for human-subjects research?

Answer: [N/A]

Justification: No new human-subjects research is conducted. The paper uses a pre-existing publicly available dataset (UTKFace).

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important component of the core methods?

    Answer: [N/A]

    Justification: LLMs are not used as a component of the core methodology. They were used only for minor writing assistance and do not impact scientific methodology, rigour, or originality.